

What chemistry faculty *believe* about doctoral education

A phenomenographic and thematic characterization

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Doctoral education produces the chemists who staff the discipline.

~2,900

chemistry PhDs
awarded each year

~196

U.S. doctoral
programs

decades

one structure,
largely unchanged

The same record documents its strains — competing responsibilities, no standard assessment of outcomes, uneven implementation, inconsistent mentorship.

Donkor & Harshman, 2023 · Collini et al., 2025

We know what faculty believe about *teaching a class*.
We don't know what they believe about **advising a PhD**.

Classroom teaching

Beliefs are well characterized — and predict what happens in the classroom.

Gibbons · Lund & Stains · Popova · Mack & Towns

Research advising

A different practice — and beliefs are tied to the practice they concern. Never characterized.

Kagan, 1992

Donkor et al. (2024) set faculty beliefs aside — a proper investigation “*warranted its own study*.” **This is that study.**

RESEARCH QUESTION

What do chemistry faculty *believe* about doctoral education?

what it's for · how students learn · who develops which skill · the structure they work in

Reforms change the structures of schooling without changing what happens inside them.

general
context of reform

the teacher's
personal profile

structural &
cultural context

teacher thinking
→ beliefs

All four shape whether a reform reaches *practice* — the model's ultimate test.

Beliefs are self-perpetuating — they can *absorb a reform without yielding to it.*

A **proposition a person holds true** and uses to interpret experience and guide action.

Rokeach, 1968 · Green, 1971 · Nespor, 1987 · Pajares, 1992

A unit of talk counts as a belief only when it is all four:

- | | | |
|----------|----------------------|--|
| 1 | Propositional | a proposition the person holds to be true |
| 2 | Evaluative | it guides how they interpret experience and decide |
| 3 | Personal | a personal stance, not a claim needing group consensus |
| 4 | Resistant | it holds up against their own counter-evidence |

An interpretive design built to capture the *range* of faculty belief

Twenty faculty

across U.S. doctoral programs — research-active, currently advising a PhD student.

Interviews + a card sort

~60-minute semi-structured interviews (a modified Teacher Beliefs Interview); the sort as a second elicitation.

Phenomenography + thematic analysis

the range of ways faculty understand doctoral education, with keyness — not frequency — as the reporting rule.

990 sub-codes



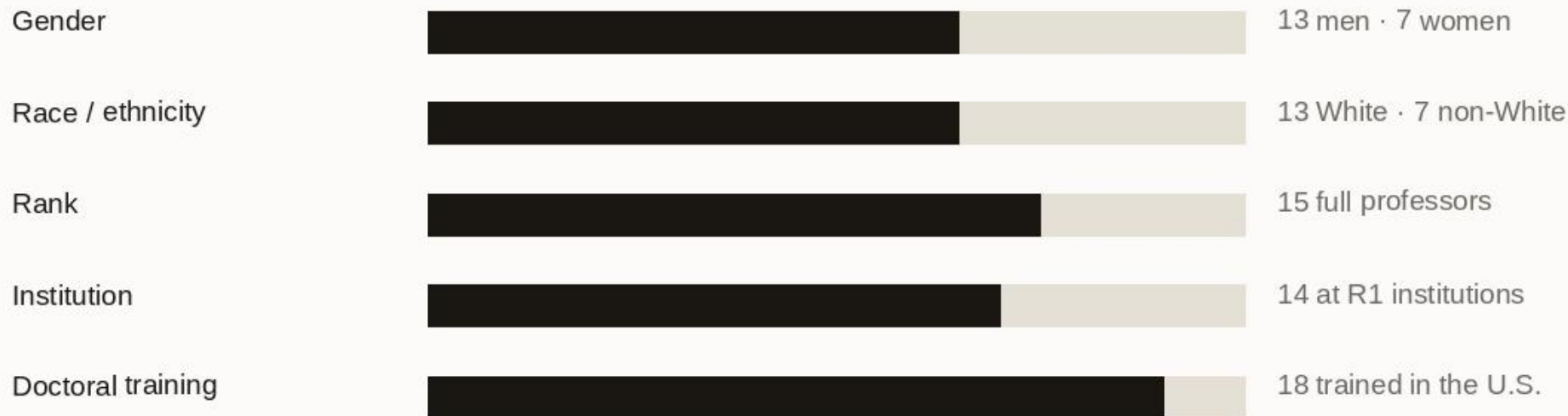
15 codes



4 themes

no new codes after interview 7

Recruited for *maximum variation* across profile and subfield



SUBDISCIPLINES

Inorganic (5) · Chemistry education (2) · Materials (2) · Physical (2) · Biophysical (2) · Organic (2) · + five singletons

Analytical generalization and transferability, not population estimates; the sample skews senior and male — see limitations.

THE FINDING

Faculty *agree* on the propositions.

They *divide* on where the boundaries fall.

“What differs is not which beliefs faculty hold, but where they draw the lines those beliefs require.”

The doctorate develops a **self-directed scientist**
in service of the student's own goals.

When they start bringing ideas forward without necessarily being prompted.

— P2, on what independence looks like

WHERE THEY DIVIDE

answering existing questions

asking new questions

The realistic goal versus the higher bar for the degree.

IC · UN · SG · FE

Students learn by doing **scaffolded research**
— and learning shows in performance, not grades.

I do. We do. You do — spread out over X number of years.

— **P5**, on how research is learned

WHERE THEY DIVIDE

unsupported struggle helps

struggle should be scaffolded

Same formative deprivation, opposite lessons — P13 vs. P7.

LE · LO · AV

Development is a **shared responsibility**
in a two-way relationship — and who develops which skill varies.

I go from the leader to the follower, and they start as the follower and end up as the leader.

— P6, on how responsibility transfers

WHERE THEY DIVIDE

faculty's to teach

student's to build

The same skill — creativity — is one faculty's to teach (P5, P17), another's for the student (P9, P11).

MB · FA · OE · RS · SD · MP

Structure shapes advising — and the **gaps**
it leaves are the program's to fix, not the student's.

Those professional skills often get kind of forgotten.

— **P2**, on the program's gap

WHERE THEY DIVIDE

the evolved system works

curricula & funding need change

How much of the structure deserves changing — P14 vs. P18.

PS · GA

Within each shared theme, faculty differ along the axes below.
Each axis names the two poles of variation; nothing is plotted on it.



THE FULL OUTCOME SPACE

Four shared themes.

Fifteen codes.

An *axis of division*

in every one.

The agreement is real — and so is the disagreement one level down. Every faculty member lands somewhere on each axis.

A **belief baseline** for changing doctoral education

1 **Content for the model**

Gives the Teacher-Centered Systemic Reform model the doctoral-education content it lacked — the shared ground a reform can assume, and the fault lines it must engage.

2 **Nests with prior work**

The goals record and the challenge record now sit beside what faculty believe. Three answers to one enterprise — they nest, not compete.

3 **Conviction vs. practice**

Thirteen of twenty hold stable convictions but refine implementation with experience — so supports aimed at implementation may fit faculty best.

IMPLICATIONS

Two levels, two lessons.

SHARED PROPOSITIONS

Common ground a reform can take as given — arguing from where faculty already stand.

BOUNDARY DISAGREEMENTS

The harder, more useful target — where colleagues who sound aligned *quietly diverge*. Surface them; don't assume them.



Thank you.

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